

Lie Theory: Chapter 4 Homework

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1. Let L_1 and L_2 be Lie algebras, and let $\varphi : L_1 \rightarrow L_2$ be a surjective Lie algebra homomorphism. True or false:

(a) $\varphi(L'_1) = L'_2$

(b) $\varphi(Z(L_1)) = Z(L_2)$

Proof.

(a) Yes. We have

$$\varphi\left(\sum a_i[x_i, y_i]\right) = \sum a_i\varphi([x_i, y_i]) = \sum a_i[\varphi(x_i), \varphi(y_i)] \in L'_2,$$

so $\varphi(L'_1) \subseteq L'_2$.

On the other hand, for any $x, y \in L_2$, we have $x', y' \in L_1$ such that $\varphi(x') = x$ and $\varphi(y') = y$, so

$$\begin{aligned}\sum a_i[x_i, y_i] &= \sum a_i[\varphi(x'_i), \varphi(y'_i)] \\ &= \sum a_i\varphi([x'_i, y'_i]) \\ &= \varphi\left(\sum a_i[x'_i, y'_i]\right) \in \varphi(L'_1),\end{aligned}$$

so $L'_2 \subseteq \varphi(L'_1)$, so $\varphi(L'_1) = L'_2$.

(b) If $x \in Z(L_1)$, then given $y \in L_2$, we have $y = \varphi(y')$ for some $y' \in L_1$, so

$$\begin{aligned}[\varphi(x), y] &= [\varphi(x), \varphi(y')] \\ &= \varphi([x, y']) \\ &= \varphi(0) \\ &= 0,\end{aligned}$$

so we have $\varphi(x) \in Z(L_2)$, and thus $\varphi(Z(L_1)) \subseteq Z(L_2)$.

The other direction does not hold, however. Consider the two-dimensional nonabelian Lie algebra $L_1 = \text{span}\{x, y\}$ with $[x, y] = x$. Consider the quotient map $\varphi : L_1 \rightarrow L_1/\text{span}\{x\}$, so $L_2 = L_1/\text{span}\{x\}$. Then $Z(L_1) = 0$, so $\varphi(Z(L_1)) = 0$, but L_2 is abelian since it is one-dimensional, so $Z(L_2) = L_2$. Thus $Z(L_2) \not\subseteq \varphi(Z(L_1))$ in general.

□

2. Use Lemma 4.4 to show that if L is a Lie algebra then L is solvable if and only if $\text{ad } L$ is a solvable subalgebra of $\mathfrak{gl}(L)$. Show that this result also holds if we replace “solvable” with “nilpotent.”

Proof. Note that since $\text{ad} : L \rightarrow \mathfrak{gl}(L)$ has kernel $Z(L)$, $\text{ad } L \cong L/Z(L)$. Lemma 4.4(a) then tells us that as a homomorphic image of L , $\text{ad } L$ is solvable if L is solvable. We then see that since $Z(L)$ is itself a solvable ideal since $Z(L)^2$, if $\text{ad } L \cong L/Z(L)$ is solvable, Lemma 4.4(b) tells us that L is solvable.

For the nilpotent versions, note that since $\text{ad} : L \rightarrow \mathfrak{gl}(L)$ is a homomorphism, we have $(\text{ad } L)^n = \text{ad}(L^n)$ for any n . Thus if L is nilpotent with $L^m = 0$, we have

$$(\text{ad } L)^m = \text{ad}(L^m) = \text{ad } 0 = 0,$$

and if instead $\text{ad } L$ is nilpotent with $(\text{ad } L)^m = 0$, we have $\text{ad}(L^m) = 0$, implying that $L^m \subseteq Z(L)$, so $L^{m+1} = 0$ and L is nilpotent. $\square$

3. Let $L = \mathfrak{n}(n, F)$. Show that L^k has a basis consisting of all the matrix units e_{ij} with $j - i > k$. Hence show that L is nilpotent. What is the smallest m such that $L^m = 0$?

Proof. As we have seen, when we bracket two elements of L together, the bottommost diagonal line of entries with nonzero elements becomes all-zero. Thus L^1 consists of matrices which are strictly upper-triangular and have zeroes in entries i, j where $j - i > 1$. Similarly, L^k has zeroes everywhere except in entries i, j where $j - i > k$. Thus it has a basis of such entries. If $\dim L = n$, then L^n would need to have a basis of matrix elements e_{ij} where $j - i > n$, but this is impossible, so $L^n = 0$ and L is nilpotent. The lowest such m is in fact $m = n - 1$, since we need the top-right entry at $1, n$ to be zero. $\square$

4. Let $L = \mathfrak{b}(n, F)$ be the Lie algebra of upper triangular $n \times n$ matrices over a field F .

(i) Show that $L' = \mathfrak{n}(n, F)$.

(ii) More generally, show that $L^{(k)}$ has a basis consisting of all the matrix units e_{ij} with $j - i \geq 2^{k-1}$. (The commutator formula for the e_{ij} given in §1.2 will be helpful.)

(iii) Hence show that L is solvable. What is the smallest m such that $L^{(m)} = 0$?

(iv) Show that if $n \geq 2$ then L is not nilpotent.

Proof.

(i) When matrix-multiplying two upper triangular matrices, the diagonal elements multiply together (i.e. $(AB)_{k,k} = A_{k,k}B_{k,k}$). Thus when we take the bracket, we get

$$\begin{aligned} [A, B]_{k,k} &= (AB - BA)_{k,k} \\ &= (AB)_{k,k} - (BA)_{k,k} \\ &= A_{k,k}B_{k,k} - B_{k,k}A_{k,k} \\ &= 0, \end{aligned}$$

so there are zeroes along the diagonal. We can still get nonzero entries above the diagonal though, so indeed $L' = \mathfrak{n}(n, F)$.

- (ii) The aforementioned commutator formula tells us that $[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{il}e_{kj}$. If $j = k$ and $i = l$ (so that the deltas are valued at 1) and $i = k$ and $l = j$ (so that $e_{il} = e_{kj}$), then $[e_{ij}, e_{kl}] = 0$, but in this case $i = j = k = l$, so we are just talking about the bracket of an element with itself. The other way $[e_{ij}, e_{kl}] = 0$ is if $j \neq k$ and $i \neq l$ so that both deltas are zero. □

5. Show that a Lie algebra is semisimple if and only if it has no non-zero abelian ideals.

Proof. Let L be a Lie algebra. First, since every abelian Lie algebra is solvable, every abelian ideal is solvable, so if there are nonzero abelian ideals of L , then L is not semisimple. Now if L is not semisimple, then let I be a nonzero solvable ideal of L and let m be the smallest integer such that $I^{(m)} = 0$. Then $I^{(m-1)} \neq 0$ and $(I^{(m-1)})' = 0$, so $I^{(m-1)}$ is abelian, but it is also an ideal and nonzero. □

7. Let L_1 and L_2 be Lie algebras, and let $\varphi : L_1 \rightarrow L_2$ be a surjective Lie algebra homomorphism. Show that if $h \in L_1$ and $\text{ad } h$ is diagonalizable then $\text{ad } \varphi(h)$ is diagonalizable.

Proof. Let $x_1, \dots, x_n$ be a basis for L_1 consisting of eigenvectors of $\text{ad } h$ and let $\lambda_1, \dots, \lambda_n$ be their corresponding eigenvalues, so that $[h, x_i] = \lambda_i x_i$ for each i . Then for each i ,

$$\begin{aligned} [\varphi(h), \varphi(x_i)] &= \varphi([h, x_i]) \\ &= \varphi(\lambda_i x_i) \\ &= \lambda_i \varphi(x_i), \end{aligned}$$

so $\varphi(x_i) = 0$ or $\varphi(x_i)$ is an eigenvector of $\text{ad } \varphi(h)$. We know that $\{\varphi(x_1), \dots, \varphi(x_n)\}$ does span L_2 since $\{x_1, \dots, x_n\}$ spans L_1 and φ is surjective, so it must contain a basis for L_2 , in particular a basis of eigenvectors, and thus $\text{ad } \varphi(h)$ is diagonalizable. □

8. Prove directly that $\mathfrak{sl}(n, \mathbb{C})$ is a simple Lie algebra for $n \geq 2$.

Proof. Let I be a nonzero ideal of $\mathfrak{sl}(n, \mathbb{C})$, and let $x \in I$ such that $x \neq 0$. If x is diagonal, then the trace is zero so not all of its diagonal entries can be equal, so let $i \neq j$ such that $x_{ii} \neq x_{jj}$ and both are nonzero. Then

$$\begin{aligned} [x, e_{ij}] &= xe_{ij} - e_{ij}x \\ &= x_{ii}e_{ij} - e_{ij}x_{jj} \\ &= (x_{ii} - x_{jj})e_{ij} \in I, \end{aligned}$$

so since a multiple of e_{ij} is in I , e_{ij} itself is in I . If x is not diagonal, then there exists a nonzero entry x_{ji} where $j \neq i$, and we see that

$$\begin{aligned} [[x, e_{ij}], e_{ij}] &= [x, e_{ij}]e_{ij} - e_{ij}[x, e_{ij}] \\ &= (xe_{ij} - e_{ij}x)e_{ij} - e_{ij}(xe_{ij} - e_{ij}x) \\ &= xe_{ij}e_{ij} - e_{ij}xe_{ij} - e_{ij}xe_{ij} + e_{ij}e_{ij}x \\ &= -2e_{ij}xe_{ij} \quad (\text{since } e_{ij}^2 = 0) \\ &= -2x_{ji}e_{ij} \in I, \end{aligned}$$

so we again have $e_{ij} \in I$. Either way, we have established $i \neq j$ such that $e_{ij} \in I$.

Now we see that $[e_{ij}, e_{kl}] \in I$ for any k and l , so using the commutator formula

$$[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{li}e_{kj},$$

so $[e_{ij}, e_{jl}] = e_{il} \in I$ for any l so long as $l \neq i$, and $[e_{ij}, e_{ki}] = -e_{kj} \in I \implies e_{kj} \in I$ for any k where $k \neq j$. From there, we use similar tricks to show that every off-diagonal matrix unit e_{kl} is in I . Finally, now that we see that $e_{m,m+1} \in I$ whenever $m < n$, we have $[e_{m,m+1}, e_{m+1,m}] = e_{m,m} - e_{m+1,m+1} \in I$. We have now shown that every standard basis element of $\mathfrak{sl}(n, \mathbb{C})$ is in I , so $I = \mathfrak{sl}(n, \mathbb{C})$, proving that $\mathfrak{sl}(n, \mathbb{C})$ is simple. $\square$

9.

- (i) Show that $\det(I + \varepsilon X)$ is a polynomial in ε of degree at most n with the first two terms

$$\det(I + \varepsilon X) = 1 + (\operatorname{tr} X)\varepsilon + \cdots.$$

If we neglect all powers of ε except 1 and ε , then we obtain the statement

$$I + \varepsilon X \in \operatorname{SL}(n, \mathbb{C}) \iff X \in \mathfrak{sl}(n, \mathbb{C}).$$

- (ii) (a) Let S be an $n \times n$ matrix. Let $(-, -)$ denote the complex bilinear form with matrix S . Show that if we let $G_S(n, \mathbb{C})$ be the set of invertible matrices A such that $(Au, Av) = (u, v)$ for all $u, v \in \mathbb{C}^n$, then $G_S(n, \mathbb{C})$ is a group.
- (b) Show that if we perform the construction in part (i) with $G_S(n, \mathbb{C})$ in place of $\operatorname{SL}(n, \mathbb{C})$, we obtain $\mathfrak{gl}_S(n, \mathbb{C})$.
- (iii) (a) An invertible matrix A is customarily said to be *orthogonal* if $A^t A = A A^t = I$; that is, if $A^{-1} = A^t$. Show that the set of $n \times n$ orthogonal matrices with coefficients in $\mathbb{C}$ is the group $G_I(n, \mathbb{C})$ and that the associated Lie algebra, $\mathfrak{gl}_I(n, \mathbb{C})$, is the space of all anti-symmetric matrices.
- (b) Prove that $\mathfrak{gl}_I(n, \mathbb{C}) \cong \mathfrak{so}(n, \mathbb{C})$. *Hint:* Use Exercise 2.11.
- (iv) A bilinear form on a vector space V is said to be *symplectic* if $(v, v) = 0$ for all $v \in V$. Show that

$$S = \begin{pmatrix} 0 & I_\ell \\ -I_\ell & 0 \end{pmatrix}$$

is the matrix of a non-degenerate symplectic bilinear form on a 2ℓ -dimensional space. The associated Lie algebra is $\mathfrak{gl}_S(2\ell, \mathbb{C}) = \mathfrak{sp}(2\ell, \mathbb{C})$.

Proof.

- (i) If λ is an eigenvalue of X and x is an eigenvector corresponding to λ , then

$$(I + \varepsilon X)x = x + \varepsilon \lambda x = (1 + \varepsilon \lambda)x,$$

so $1 + \varepsilon\lambda$ is an eigenvalue of $I + \varepsilon X$. Thus if $\lambda_1, \dots, \lambda_n$ are the eigenvalues of X repeated according to multiplicity, then

$$\det(I + \varepsilon X) = \prod_{1 \leq i \leq n} (1 + \varepsilon\lambda_i),$$

which is indeed a polynomial of degree n in ε whose first term is 1 and whose second term is

$$\sum_i \varepsilon\lambda_i = (\operatorname{tr} X)\varepsilon.$$

□